

Universal Charter of the Open Physics Research Collective (OPRC)

The Founding Assembly

May 3, 2026

1 Section I: Foundational Principles

1.1 Sub-section 1.1: Legal Status and Digital Existence

1.1.1 Article 1.1.1: The Unincorporated Association Mandate

1. Sub-article 1.1.1.1: Voluntary Nature

- (a) **Clause:** The OPRC shall exist solely as a voluntary, unincorporated association of global researchers.
 - Compliance with digital jurisdiction.
 - Non-recognition of corporate personhood.
 - Autonomy of individual contributors.
 - Responsibility for local legal compliance.
 - Protection of the collective's decentralized identity.
- (b) **Clause:** Liability is restricted to individual actions and not the collective body.
 - Indemnification protocols for peer reviewers.
 - Standard of care for theoretical proofs.
 - Limitations on digital asset liability.
 - Arbitration through community consensus.
 - Protection of the OPRC brand and name.

2. Sub-article 1.1.1.2: Digital-First Infrastructure

- (a) **Clause:** The primary residence of the OPRC is the distributed digital ledger and version control systems.
 - Maintenance of the 'Genesis' repository.
 - Protocol for redundant data storage.
 - Uptime requirements for communication hubs.
 - Accessibility standards for low-bandwidth regions.
 - Cryptographic verification of the Charter's integrity.

1.2 Sub-section 1.2: Radical Transparency and Mission

1.2.1 Article 1.2.1: The Universal Inquiry Protocol

1. Sub-article 1.2.1.1: All-Fields Mandate

- (a) **Clause:** Research shall span High-Energy, Quantum, Materials, and Biophysics.
 - Cross-pollination between string theory and condensed matter.
 - Standardization of mathematical notation.
 - Requirement for field-agnostic summaries.
 - Integration of gravitational wave data.
 - Protocols for interdisciplinary peer review.

2 Section II: Governance and Decision Making

2.1 Sub-section 2.1: Consensus and Rough Consensus Models

2.1.1 Article 2.1.1: The RFC (Request for Comments) Standard

1. Sub-article 2.1.1.1: Submission and Period of Review

- (a) **Clause:** All major research initiatives must be prefaced by a public RFC document.
 - Minimum review period of fourteen (14) digital days.
 - Requirement for technical abstract and computational requirements.
 - Tagging of relevant research clusters.
 - Public archival of all comments and rebuttals.
 - Protocol for 'Hot-Fix' RFCs in case of critical theoretical errors.
- (b) **Clause:** Decision making follows the 'Rough Consensus and Working Code' philosophy.
 - Identification of blocking technical objections.
 - Neutralization of non-technical grievances.
 - Role of the 'Shepherd' in distilling consensus.
 - Resolution of 'Deadlock' scenarios through computational proof.
 - Periodic re-evaluation of 'Settled' RFCs.

2.2 Sub-section 2.2: Distributed Leadership and Roles

2.2.1 Article 2.2.1: Domain Maintainers and Cluster Leads

1. Sub-article 2.2.1.1: Appointment and Responsibilities

- (a) **Clause:** Cluster Leads are elected based on contribution volume and peer recognition.
 - Oversight of domain-specific peer review queues.
 - Maintenance of the cluster bibliography.
 - Facilitation of cross-cluster 'Symmetry' meetings.
 - Authority to merge approved RFCs into the main branch.
 - Term limits and succession planning protocols.

2.3 Sub-section 2.3: Dispute Resolution and Arbitration

2.3.1 Article 2.3.1: The Technical Arbitration Council

1. Sub-article 2.3.1.1: Conflict Mitigation

- (a) **Clause:** Disputes regarding theoretical priority or methodology shall be settled via Council review.
 - Random selection of five (5) disinterested peer maintainers.
 - Blind review of conflicting proofs or data.
 - Adherence to the OPRC Code of Conduct during mediation.
 - Finality of Council decisions on OPRC-branded repositories.
 - Right to 'Fork' for dissenting researchers under separate branding.

3 Section III: Research Methodology

3.1 Sub-section 3.1: Standards of Peer Review

3.1.1 Article 3.1.1: Open-Continuous Review Protocol

1. Sub-article 3.1.1.1: Radical Transparency in Critique

- (a) **Clause:** All reviews must be signed and published alongside the research findings.
 - Elimination of anonymous obstructionism.
 - Permanent archival of review history in the project branch.
 - Cross-linking of reviewer expertise to cluster tags.

- Requirement for constructive rebuttal for all objections.
- Integration with open platforms like PubPeer and arXiv.
- (b) **Clause:** The review process is iterative and never truly closed.
 - Continuous versioning of published papers based on community feedback.
 - 'Live-Paper' status for ongoing theoretical developments.
 - Protocol for post-publication error identification.
 - Recognition of significant reviewers as secondary contributors.
 - Implementation of a 'Quality Score' for verified proofs.

3.2 Sub-section 3.2: Open Data and Metadata Standards

3.2.1 Article 3.2.1: The 'Raw-to-Result' Mandate

1. Sub-article 3.2.1.1: Data Availability and Integrity

- (a) **Clause:** All experimental and simulation data must be hosted on immutable repositories.
 - Utilization of DOIs via Zenodo or similar services.
 - Mandatory submission of raw data prior to peer review.
 - Standardization of CSV/HDF5 formats for physics datasets.
 - Inclusion of sensor calibration logs for physical experiments.
 - Checksum verification for data integrity.
- (b) **Clause:** Metadata must adhere to the 'FAIR' principles.
 - Findability through standardized ontology.
 - Accessibility via open-source API protocols.
 - Interoperability across different computational physics tools.
 - Reusability through clear license tagging.
 - Implementation of the 'Physical Metadata' schema.

3.3 Sub-section 3.3: Computational Reproducibility

3.3.1 Article 3.3.1: The Containerization Requirement

1. Sub-article 3.3.1.1: Software Environment Standardization

- (a) **Clause:** Calculations must be provided in an executable, containerized format.
 - Support for Docker or Singularity images.
 - Inclusion of environment lock files (e.g., conda, poetry).
 - Documentation of hardware requirements for simulations.
 - Automated verification of results via GitHub Actions.
 - Prohibition of proprietary compiler dependencies.

4 Section IV: Intellectual Property and Licensing

4.1 Sub-section 4.1: The Open Source Mandate

4.1.1 Article 4.1.1: Licensing of Theoretical and Computational Works

1. Sub-article 4.1.1.1: Default Public Domain and Copyleft

- (a) **Clause:** All theoretical derivations and mathematical proofs shall be released under CC-BY 4.0.
 - Permanent waiver of commercial exclusivity.
 - Requirement for citation in all derivative academic works.
 - Preservation of moral rights for original authors.
 - Compatibility with LaTeX source archiving.
 - Non-revocable nature of the open license.
- (b) **Clause:** All software, scripts, and simulation code must utilize the MIT or GPLv3 licenses.

- Mandatory inclusion of license headers in all source files.
- Freedom to fork and modify for non-OPRC initiatives.
- Prohibition of 'black-box' binary distributions without source.
- Verification of license compatibility with existing libraries.
- Protection of contributors from downstream software liability.

4.2 Sub-section 4.2: Anti-Patent Policy

4.2.1 Article 4.2.1: Defense of Fundamental Physical Principles

1. Sub-article 4.2.1.1: Prohibition of Privatization

- (a) **Clause:** Members shall not seek patents on fundamental laws discovered within the OPRC.
- Automatic 'Prior Art' status for all published pre-prints.
 - Commitment to challenge predatory patents on physical constants.
 - Use of the 'Defensive Patent License' where applicable.
 - Community-funded legal opposition to privatization efforts.
 - Transparency requirements for individual member patent filings.

4.3 Sub-section 4.3: Attribution and Academic Credit

4.3.1 Article 4.3.1: The Contributor Covenant for Papers

1. Sub-article 4.3.1.1: Recognition of Non-Traditional Contributions

- (a) **Clause:** Credit shall be extended to coders, data maintainers, and reviewers.
- Standardized 'CONTRIBUTORS.md' in every research repository.
 - Use of the CRediT (Contributor Roles Taxonomy) system.
 - ORCID integration for all participating members.
 - Fractional authorship models based on commit history.
 - Protection against 'honorary' or 'gift' authorship.

5 Section V: Membership and Community

5.1 Sub-section 5.1: Onboarding and Contribution

5.1.1 Article 5.1.1: The Global Researcher Access Protocol

1. Sub-article 5.1.1.1: Criteria for Entry

- (a) **Clause:** Membership is open to any individual demonstrating commitment to the scientific method.
- Verification through existing research profile or public repository.
 - Acceptance of the OPRC Open Source Mandate.
 - Successful completion of the 'Initial Contribution' period.
 - Digital signature of the Founding Charter.
 - Declaration of institutional or commercial conflicts of interest.
- (b) **Clause:** The OPRC maintains a policy of non-discrimination based on formal affiliation.
- Inclusion of independent scholars and citizen scientists.
 - Support for student-led research initiatives.
 - Cross-border collaboration regardless of political boundaries.
 - Provisions for low-resource environment participation.
 - Protections for minor researchers under supervision.

5.2 Sub-section 5.2: Code of Conduct and Ethics

5.2.1 Article 5.2.1: Standards of Collaborative Discourse

1. Sub-article 5.2.1.1: The 'Open Science' Ethics Mandate

- (a) **Clause:** All members shall adhere to the Contributor Covenant and OPRC-specific ethics.
 - Prohibition of scientific fraud or data manipulation.
 - Commitment to professional civility during RFC reviews.
 - Zero-tolerance for harassment or exclusionary behavior.
 - Transparency in funding and external influence.
 - Obligation to report observed theoretical errors in others' work.

5.3 Sub-section 5.3: Contributor Tiers and Meritocracy

5.3.1 Article 5.3.1: Defining the Contribution Hierarchy

1. Sub-article 5.3.1.1: Tiered Responsibilities

- (a) **Clause:** Roles are distinguished by contribution volume and community trust.
 - **Observer:** Access to public documentation and discussions.
 - **Contributor:** Authorized to submit PRs and RFC comments.
 - **Maintainer:** Vested with repo merge authority and RFC shepherding.
 - **Cluster Lead:** Responsible for high-level research domain direction.
 - **Emeritus:** Recognition for long-term historical contributions.

6 Section VI: Resource Management

6.1 Sub-section 6.1: Digital Infrastructure and Stability

6.1.1 Article 6.1.1: Management of Shared Computational Assets

1. Sub-article 6.1.1.1: Resource Allocation Protocols

- (a) **Clause:** Shared high-performance computing (HPC) resources shall be allocated based on RFC priority.
 - Preference for projects with high interdisciplinary impact.
 - Requirement for efficient code audits prior to large-scale runs.
 - Monitoring of energy consumption and computational waste.
 - Fair-use limits for individual researcher nodes.
 - Protocol for 'Emergency Compute' during critical experimental windows.
- (b) **Clause:** The OPRC shall maintain a decentralized grid of contributor-provided hardware.
 - Security standards for volunteer-run computational nodes.
 - Incentivization through 'Computation Credits' for hardware providers.
 - Standardization of distributed task scheduling (e.g., BOINC or custom).
 - Data privacy safeguards for sensitive experimental telemetry.
 - Redundancy requirements for mission-critical simulations.

6.2 Sub-section 6.2: Funding and Donation Ethics

6.2.1 Article 6.2.1: The Transparent Treasury Standard

1. Sub-article 6.2.1.1: External Financial Contributions

- (a) **Clause:** The OPRC accepts donations solely to maintain infrastructure and open-access fees.
 - Prohibition of 'directed' research funding that compromises neutrality.
 - Public disclosure of all individual donations exceeding baseline limits.
 - Utilization of crypto-treasuries or open-collective platforms for transparency.
 - Annual auditing of the collective's digital wallet by non-leads.
 - Clause for the rejection of funds from unethical industry sources.

6.3 Sub-section 6.3: Digital Archiving and Persistence

6.3.1 Article 6.3.1: The 'Perma-Archive' Mandate

1. Sub-article 6.3.1.1: Long-term Proof Preservation

- (a) **Clause:** All finalized research must be mirrored across at least three geographic zones.
 - Integration with the Software Heritage archive.
 - Periodic 'freeze' of core repositories for offline storage.
 - Maintenance of a decentralized IPFS mirror for OPRC papers.
 - Protocol for hardware-agnostic documentation formats.
 - Succession plan for domain names and digital identity keys.

7 Section VII: External Relations

7.1 Sub-section 7.1: Academic and Institutional Interaction

7.1.1 Article 7.1.1: The Bridge-Protocol for Universities

1. Sub-article 7.1.1.1: Collaborative Frameworks

- (a) **Clause:** The OPRC shall facilitate joint research efforts with established academic institutions.
 - Protocol for dual-affiliation credit.
 - Management of shared experimental facility access.
 - Standardization of cross-institutional data transfer.
 - Alignment with university open-access mandates.
 - Safeguards for student researchers' primary academic standing.
- (b) **Clause:** Interaction with traditional journals remains secondary to open-first dissemination.
 - Encouragement of 'Overlay Journal' models.
 - Preference for journals that allow immediate pre-print archiving.
 - Collective negotiation for reduced Article Processing Charges (APCs).
 - Preservation of copyright within the collective for all external publications.
 - Protocol for responding to predatory journal solicitations.

7.2 Sub-section 7.2: Public Outreach and Education

7.2.1 Article 7.2.1: The 'Physics for All' Initiative

1. Sub-article 7.2.1.1: Science Communication Standards

- (a) **Clause:** Finalized research must include a 'Layperson's Abstract' to maximize accessibility.
 - Translation of core findings into non-technical languages.
 - Development of open-source educational visualizations.
 - Facilitation of public 'Ask Me Anything' (AMA) sessions for new theories.
 - Engagement with citizen science platforms for data collection.
 - Protection of scientific accuracy over sensationalism.

7.3 Sub-section 7.3: Industry and Commercial Interaction

7.3.1 Article 7.3.1: The 'Neutral-Ground' Policy

1. Sub-article 7.3.1.1: Commercial Use of Open Findings

- (a) **Clause:** The OPRC remains platform-neutral and industrially agnostic.
 - Right of commercial entities to use OPRC software as per MIT/GPL licenses.
 - Prohibition of industry-funded 'Embargo Periods' for research.
 - Protocol for industry researchers to contribute as individuals.
 - Refusal of any partnership requiring non-disclosure agreements (NDAs).
 - Transparency of industry-affiliated contributors' professional roles.

8 Section VIII: Domain-Specific Operational Mandates

8.1 Sub-section 8.1: High-Energy and Particle Physics (HEPP)

8.1.1 Article 8.1.1: Large-Scale Dataset Ingestion

1. Sub-article 8.1.1.1: Collider Data and Phenomenology

- (a) **Clause:** HEPP projects must prioritize the ingestion of open-access collider data.
 - Protocol for interfacing with CERN Open Data portal.
 - Requirements for standardized Monte Carlo simulation event formats.
 - Verification of kinematic reconstructions by secondary clusters.
 - Inclusion of systematic uncertainty covariance matrices.
 - Open-source implementation of custom jet-clustering algorithms.
- (b) **Clause:** Theoretical phenomenology must remain verifiable via fast-sim tools.
 - Requirement for Pythia/MadGraph configuration file disclosure.
 - Benchmarking against the Standard Model baseline.
 - Mandatory sensitivity analysis for Beyond Standard Model (BSM) searches.
 - Protocol for tagging 'anomalous' events for community vetting.
 - Preservation of 'Null Result' datasets for future meta-analysis.

8.2 Sub-section 8.2: Quantum Information and Matter

8.2.1 Article 8.2.1: Quantum Computational Physics

1. Sub-article 8.2.1.1: Circuit Transparency and Verification

- (a) **Clause:** All quantum algorithms must be provided in OpenQASM or equivalent code.
 - Requirement for noise-model parameters in simulations.
 - Disclosure of error-correction overheads for proposed gates.
 - Benchmarking of quantum-classical hybrid variational solvers.
 - Verification of entanglement entropy proofs via open scripts.
 - Accessibility of pulse-level control data for hardware experiments.
- (b) **Clause:** Condensed matter simulations must adhere to the 'Many-Body' standard.
 - Standardization of tensor network contraction protocols.
 - Mandatory convergence testing for Density Functional Theory (DFT).
 - Protocol for sharing Wannier function datasets.
 - Requirement for pseudopotential file availability.
 - Documentation of thermodynamic limit extrapolations.

9 Section IX: Constitutional Evolution and Legal Defense

9.1 Sub-section 9.1: Amendment Protocols

9.1.1 Article 9.1.1: The Living Charter Standard

1. Sub-article 9.1.1.1: Proposal and Ratification

- (a) **Clause:** Any Maintainer or Lead may propose a 'Constitutional RFC' to amend this Charter.
 - Minimum ninety (90) day community deliberation period.
 - Requirement for a 'Rough Consensus' check across all research clusters.
 - Mandatory impact assessment on existing open-source licenses.
 - Preservation of the 'Preamble' as an unalterable core principle.
 - Cryptographic timestamping of all approved versions.
- (b) **Clause:** Amendments must maintain backward compatibility with previous research grants.
 - Protection of 'Legacy' project status under prior rules.

- Identification of conflicting clauses in proposed revisions.
- Inclusion of 'Grandfather' clauses for ongoing physical experiments.
- Documentation of the rationale for every line-item change.
- Right of members to 'opt-out' of new governance via project migration.

9.2 Sub-section 9.2: Collective Dissolution and Asset Transition

9.2.1 Article 9.2.1: The Graceful Exit Protocol

1. Sub-article 9.2.1.1: Winding Down Operations

- (a) **Clause:** Dissolution occurs only upon a clear failure of consensus or digital infrastructure.
- Automatic release of all private branch data into the public domain.
 - Transition of digital assets (domains, servers) to a non-profit foundation.
 - Formal notice to all academic and institutional partners.
 - Archival of the 'Genesis' repository on permanent physical media.
 - Settlement of all outstanding 'Computation Credits.'

9.3 Sub-section 9.3: Legal Defense and Indemnification

9.3.1 Article 9.3.1: The Protective Shield Standard

1. Sub-article 9.3.1.1: Collective Support for Researchers

- (a) **Clause:** The OPRC shall provide documentation and prior art for members facing patent litigation.
- Maintenance of a 'Defense Repository' of open-source proofs.
 - Expert testimony from cluster leads in defense of open inquiry.
 - Leveraging of the 'Anti-Patent Policy' in legal challenges.
 - Coordination with global digital rights organizations (e.g., EFF).
 - Limitation of collective liability for individual member legal costs.

10 Section X: Final Ratification and Persistent Inquiry

10.1 Sub-section 10.1: Ratification and Initial Versioning

10.1.1 Article 10.1.1: The Genesis Commit

1. Sub-article 10.1.1.1: Formal Commencement

- (a) **Clause:** This Charter is formally ratified upon its initial commit to the primary OPRC repository.
- Requirement for cryptographic signatures from at least three (3) Founding Assembly members.
 - Immediate transition from 'Draft' to 'Live' status upon timestamp verification.
 - Public broadcast of the Charter hash across all OPRC communication channels.
 - Archival of the genesis document in multiple decentralized storage clusters.
 - Activation of all Article IV open-source mandates for existing projects.
- (b) **Clause:** Initial Cluster Leads shall be appointed as 'Interim Domain Maintainers.'
- Responsibility for establishing domain-specific contribution guides.
 - Setup of initial CI/CD pipelines for computational verification.
 - Mapping of existing open-access physics datasets for ingestion.
 - Facilitation of the first OPRC Global Synchrony meeting.
 - Drafting of the 'Member Onboarding Handbook' as a supplementary guide.

We, the Founding Assembly, hereby establish this framework as a testament to the fact that physical knowledge belongs to the collective mind of humanity.

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